

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 7-2023: A 70-Year-Old Man with Covid-19, Respiratory Failure, and Rashes

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PRESENTATION OF CASE

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Dr. Carolina Geadas (Medicine): A 70-year-old man with a history of kidney transplantation was evaluated at this hospital because of multiple erythematous rashes that had developed during an admission for respiratory failure, pneumococcal bacteremia, and kidney failure.

Approximately 4.5 years before the current admission, the patient received a diagnosis of antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis after he presented with progressive glomerulonephritis with kidney failure and new hemoptysis. The anti-myeloperoxidase antibody titer was 221.0 U (reference value, <2.8). The patient underwent plasmapheresis, and cyclophosphamide and prednisone were administered. Hemodialysis was initiated. Two weeks later, rituximab was administered, and cyclophosphamide and prednisone were discontinued. During the next 4 years, his renal function did not improve, and treatment with rituximab and hemodialysis was continued.

Four months before the current admission, the patient underwent transplantation of a kidney from a deceased donor with hepatitis C virus antibody-positive status. After the kidney transplantation, immunosuppressive therapy included a dose of basiliximab, mycophenolate mofetil, tacrolimus, and tapering doses of prednisone. The patient had delayed graft function; the blood creatinine level decreased from approximately 4.00 mg per deciliter (354 mmol per liter) immediately after transplantation to 1.50 mg per deciliter (133 mmol per liter) 6 weeks later. Six days before the current admission, at a routine follow-up visit, the creatinine level was 1.33 mg per deciliter (118 mmol per liter).

Four days before the current admission, the patient began to have central abdominal discomfort, nausea, anorexia, chills, and diarrhea with loose stools three times per day. During the next several days, he noticed decreased urine output and malaise, and he began to have rhinorrhea, cough, sinus congestion, and dyspnea. He was brought to the emergency department of this hospital.

A review of systems was notable for palpitations and dizziness and was negative for fever, emesis, and hematuria. The patient's brother had recently received a diagnosis of coronavirus disease 2019 (Covid-19). The patient's medical history was notable for vasculitis with end-stage kidney disease (culminating in kidney transplantation), coronary artery disease (with placement of a stent in the right coronary artery), atrial fibrillation, type 2 diabetes mellitus, hypertension, cholelithiasis (with cholecystectomy), colonic polyps and diverticulosis, gout, gastroesophageal reflux disease, hypothyroidism, sleep apnea, hernia (with hernia repair), and spinal stenosis. Medications taken at home included mycophenolate mofetil, tacrolimus, prednisone, trimethoprim-sulfamethoxazole, valganciclovir, warfarin, aspirin, atorvastatin, levothyroxine, and cholecalciferol, as well as lorazepam as needed. Penicillin had caused anaphylaxis.

The patient was a retired business owner and lived with family in Boston. He had a remote history of tobacco smoking that totaled 10 pack-years; he had quit smoking approximately 40 years earlier. He did not drink alcohol or use other substances. His family history was notable for diabetes, hypertension, and coronary artery disease in his father and siblings. There was no known family history of kidney or autoimmune disease.

The temperature was 36.4°C, the heart rate 130 beats per minute, the blood pressure 109/56 mm Hg, and the respiratory rate 24 breaths per minute. The oxygen saturation was 84% while the patient was breathing ambient air, 92% while he was receiving oxygen through a nasal cannula at a rate of 6 liters per minute, and 100% while he was receiving oxygen through a high-flow nasal cannula at a rate of 60 liters per minute. On examination, he appeared acutely ill and weak. He had bitemporal cachexia, dry oral mucosa, and increased work of breathing. Auscultation of the chest revealed tachypnea, diminished breath sounds and crackles in the lungs, and tachycardia. There was tenderness in the periumbilical area and the right lower quadrant, at the site of the kidney transplantation; the incision was well healed.

Blood levels of lipase, calcium, magnesium, and phosphorus were normal; other laboratory test results are shown in Table 1. Blood and

urine specimens were obtained for culture. An electrocardiogram showed possible sinus tachycardia with right bundle-branch block. Vancomycin and cefepime were administered intravenously. Three hours after presentation, a test for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) RNA was positive, and remdesivir and dexamethasone were administered intravenously.

Dr. Joseph D. King: Radiography of the chest (Fig. 1A) revealed confluent basilar consolidation in the left lung and scattered patchy opacities in the right lung. Ultrasonography that targeted the renal allograft revealed normal renal parenchyma and a trace amount of adjacent free fluid. Doppler ultrasonography (Fig. 1B through 1E) showed patent vasculature with normal flow velocities but showed elevated renal arterial resistive indexes, which are suggestive of allograft dysfunction. Computed tomography (CT) of the chest (Fig. 1F and 1G), performed without the administration of intravenous contrast material, revealed confluent consolidation in the lower lobes of both lungs, with more extensive findings in the left lung than in the right lung, as well as patchy multifocal consolidative and ground-glass opacities in the upper lobes of both lungs and the middle lobe of the right lung. Concurrent CT of the abdomen and pelvis (Fig. 1H), performed without the administration of intravenous contrast material, was remarkable for only a trace amount of free fluid in the right lower quadrant, along the renal allograft.

Dr. Geadas: The patient was admitted to the medical intensive care unit (ICU) of this hospital. Phenylephrine was administered intravenously. He could not be weaned from the high-flow nasal cannula; because he had increased work of breathing and tachypnea, tracheal intubation was performed and mechanical ventilation initiated. Bronchoscopy revealed dark secretions in the lower lobe and lingula of the left lung. Blood cultures were positive for *Streptococcus pneumoniae* in four of four bottles. A blood test for cytomegalovirus (CMV) DNA and a urine test for legionella antigen were negative, and serum levels of 1,3- β -D-glucan and galactomannan were undetectable. Azithromycin, cefepime, linezolid, amiodarone, furosemide, and vitamin K were administered intravenously, and valganciclovir and atovaquone were administered orally. A trans-

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Initial Evaluation	Hospital Day 4	Hospital Day 14
Blood				
Hemoglobin (g/dl)	13.5–17.5	13.8	10.1	7.7
Hematocrit (%)	41.0–53.0	41.5	29.8	24.0
White-cell count (per μ l)	4500–11,000	320	9520	3250
Differential count (per μ l)				
Neutrophils	1800–7700	50	8580	3050
Lymphocytes	1000–4800	60	80	30
Monocytes	200–1200	110	580	80
Eosinophils	0–900	—	0	80
Platelet count (per μ l)	150,000–400,000	168,000	106,000	35,000
Sodium (mmol/liter)	135–145	132	134	138
Potassium (mmol/liter)	3.4–5.0	3.1	4.8	4.9
Chloride (mmol/liter)	98–108	95	101	103
Carbon dioxide (mmol/liter)	23–32	19	18	18
Urea nitrogen (mg/dl)	8–25	67	31	20
Creatinine (mg/dl)	0.60–1.50	4.08	1.63	1.02
Glucose (mg/dl)	70–110	83	170	161
Albumin (g/dl)	3.3–5.0	3.5	1.8	2.3
Alanine aminotransferase (U/liter)	10–55	18	190	36
Aspartate aminotransferase (U/liter)	10–40	61	814	46
Alkaline phosphatase (U/liter)	45–115	63	134	90
Total bilirubin (mg/dl)	0.0–1.0	1.2	2.8	4.4
Direct bilirubin (mg/dl)	0.0–0.4	0.8	2.2	4.0
Lactate (mmol/liter)	0.5–2.2	1.5	2.4	2.0
Prothrombin time (sec)	11.5–14.5	56.8	21.2	18.4
International normalized ratio	0.9–1.1	6.8	1.9	1.6
Partial-thromboplastin time (sec)	22.0–36.0	—	76.8	44.6
Tacrolimus (ng/ml)	>2.0	29.9	13.1	3.2
High-sensitivity troponin T (ng/liter)	0–9	105	—	—
N-terminal pro-B-type natriuretic peptide (pg/ml)	<900	63,959	—	—
C-reactive protein (mg/liter)	0.0–8.0	120.4	—	—
Ferritin (μ g/liter)	20–300	4261	—	—
D-dimer (ng/ml)	<500	—	916	—
Arterial blood gas				
Fraction of inspired oxygen	—	—	0.30	0.55
pH	7.35–7.45	—	7.38	7.31
Partial pressure of carbon dioxide (mm Hg)	35–42	—	31	40
Partial pressure of oxygen (mm Hg)	80–100	—	80	65
Urine				
Color	Yellow	Yellow	Yellow	—
Clarity	Clear	Turbid	Slightly cloudy	—

Table 1. (Continued.)

Variable	Reference Range, Adults†	On Initial Evaluation	Hospital Day 4	Hospital Day 14
pH	5.0–9.0	5.5	6.0	—
Specific gravity	1.001–1.035	1.019	1.020	—
Glucose	Negative	Negative	Negative	—
Ketones	Negative	Negative	Negative	—
Leukocyte esterase	Negative	Negative	Negative	—
Nitrite	Negative	Negative	Negative	—
Blood	Negative	2+	3+	—
Protein	Negative	2+	2+	—
Erythrocytes (per high-power field)	0–2	10–20	20–50	—
Leukocytes (per high-power field)	<10	0	10–20	—
Bacteria	None	1+	2+	—

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for lactate to milligrams per deciliter, divide by 0.1110.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

thoracic echocardiogram showed low-normal ventricular function, atrial enlargement, and no valvular vegetations. Continuous venovenous hemofiltration was initiated. ANCA titers were undetectable.

On hospital day 3, excoriation and erythema of the buttock and a purpuric plaque on the sacrum were noted (Fig. 2A). Repeat CT of the chest, abdomen, and pelvis showed no changes. Punch biopsy of skin on the right buttock was performed. The antibiotic regimen was transitioned to intravenous ceftriaxone.

The following day, histologic examination of the skin-biopsy specimen revealed epidermal necrosis, with scattered apoptotic and necrotic keratinocytes and extravasated erythrocytes. No microorganisms were identified, and microbiologic culture of the biopsy specimen was negative. Laboratory test results obtained on hospital day 4 are shown in Table 1.

During the next 6 days, treatment with continuous venovenous hemofiltration and intravenous ceftriaxone was continued. The level of mechanical ventilation was reduced from mandatory provided breaths to spontaneous ventilation, and the mode was switched to pressure-support ventilation. Repeat blood cultures were negative. On hospital day 11, the patient's respi-

ratory status worsened; the fraction of inspired oxygen was increased, and a mandatory respiratory rate was reinstituted.

Two days later, a new well-circumscribed purpuric patch was noted on the abdomen (Fig. 2B). The patch measured 5 cm in greatest dimension and had a hyperpigmented halo but had no fluctuance or crepitus. The next day, a similar-appearing lesion was noted on the right thigh. Laboratory test results obtained on hospital day 14 are shown in Table 1.

Dr. King: Repeat CT of the chest, abdomen, and pelvis showed increased consolidative and ground-glass opacities in both lungs.

Dr. Geadas: Ceftriaxone was discontinued, and vancomycin, cefepime, and liposomal amphotericin B were administered intravenously. A diagnosis and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Michael K. Mansour: I participated in the care of this patient, and I am aware of the final diagnosis. This 70-year-old man — who had a history of ANCA-associated vasculitis that was in remission, diabetes, and recent kidney transplantation — was admitted to this hospital with SARS-CoV-2 infection and respiratory failure, for

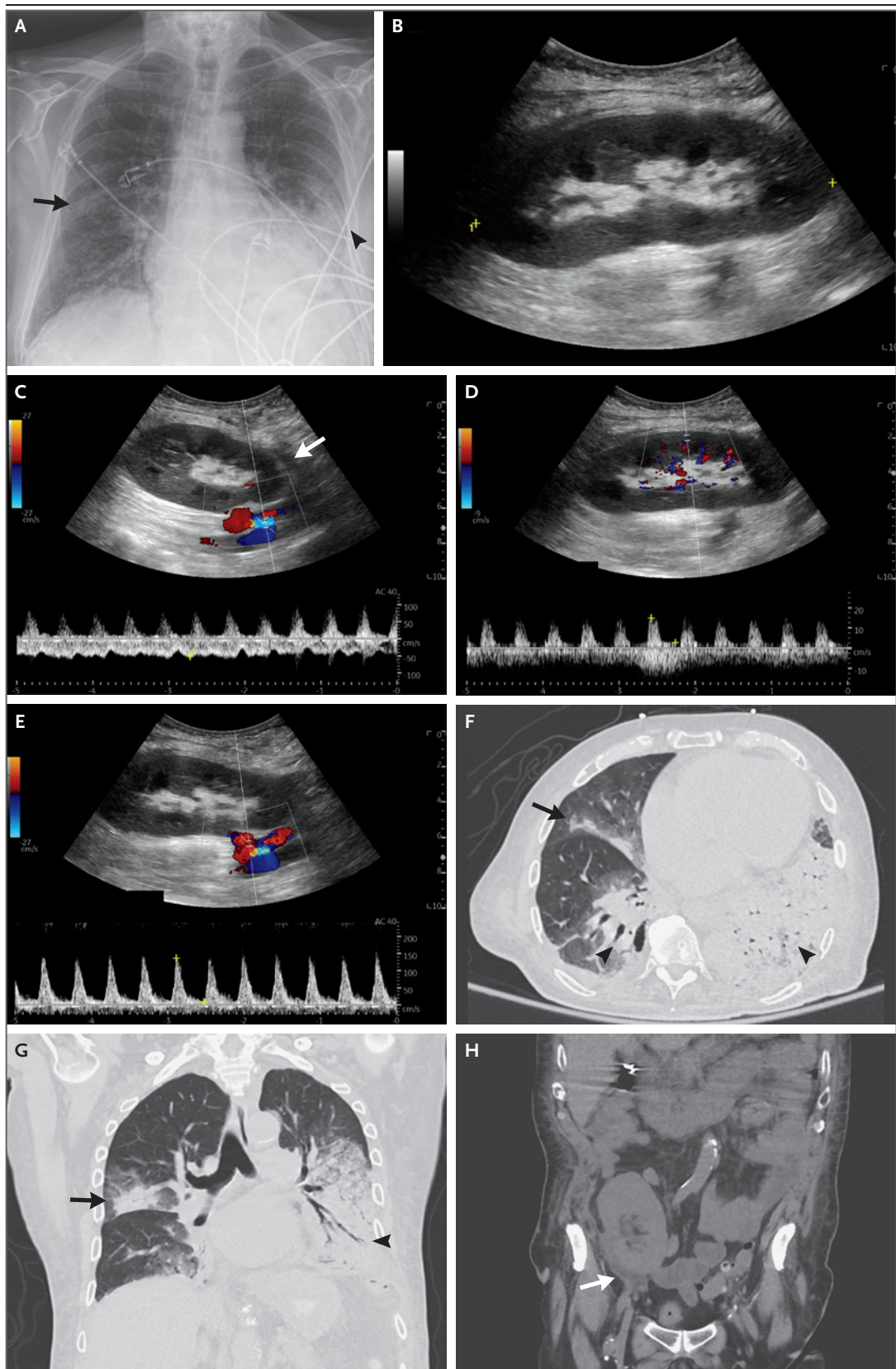


Figure 1 (facing page). Imaging Studies.

A radiograph of the chest (Panel A) shows confluent basilar consolidation in the left lung (arrowhead) and scattered patchy opacities in the right lung (arrow). Doppler ultrasound images targeting the renal allograft (Panels B through E) show normal renal parenchyma with a trace amount of perinephric fluid (Panel C, arrow) and patent vasculature with normal flow velocities but show elevated renal arterial resistive indexes, which are suggestive of allograft dysfunction. CT images of the chest (Panels F and G), obtained without the administration of intravenous contrast material, show confluent consolidation with air bronchograms in the lower lobes of both lungs (arrowheads), as well as patchy multifocal consolidative and ground-glass opacities in both upper lobes and the right middle lobe (arrows). A CT image of the abdomen and pelvis (Panel H), obtained without the administration of intravenous contrast material, shows a trace amount of free fluid in the right lower quadrant (arrow), along the renal allograft.

which he received mechanical ventilation. During the admission, multiple hemorrhagic, non-blanching skin lesions developed. When I initially evaluated this patient, I constructed a differential diagnosis on the basis of the objective findings that were associated with his immunocompromised state after solid-organ transplantation, as well as his clinical presentation in the context of Covid-19.

This patient had undergone kidney transplantation 4 months before admission, and his induction immunosuppressive therapy included basiliximab (an interleukin-2 receptor blocker), mycophenolate mofetil, tacrolimus, and prednisone. He had delayed graft function but eventually had recovery of renal function. Four days before admission, an acute illness with mainly gastrointestinal symptoms, including diarrhea and chills, developed. Covid-19 had been diagnosed in his brother, so there was a known sick contact. Two days before admission, an upper respiratory illness developed, with cough and rhinorrhea that progressed to dyspnea. The patient presented to the emergency department with a low oxygen saturation (84%) and increased work of breathing. Covid-19 and *S. pneumoniae* bacteremia were diagnosed. Subsequently, hemorrhagic rashes involving multiple areas of the body developed. Given the nature and rapid onset of the skin lesions, infectious causes rose to the top of the differential diagnosis. Noninfectious causes, including drug allergy, autoimmunity, mechanical injury, and cancer, were also

**Figure 2. Clinical Photographs.**

Panel A shows a violaceous purpuric plaque on the sacrum with color change at the midline, which was noted on hospital day 3. Panel B shows a lavender purpuric patch on the left side of the lower abdomen with a surrounding pink macular halo, which was noted on hospital day 13.

considered. Because the patient was in an immunocompromised state, I will explore potential infectious processes that occur in association with immunosuppression, critical illness, and rapidly disseminated skin lesions.

VIRAL INFECTIONS

Rash is a characteristic feature of viral infection. Reactivation of herpesviruses is common in immunocompromised hosts. Specifically, reactiva-

tion of herpes simplex virus, CMV, and Epstein–Barr virus can occur after transplantation, and such infections can have cutaneous manifestations. However, the related skin lesions are typically maculopapular and not hemorrhagic. Varicella–zoster virus infection occurs at a higher prevalence among transplant recipients than in the general population, but the rash is vesicular in nature.¹ Infection with viruses other than herpesviruses, such as parvovirus and human immunodeficiency virus, can also have cutaneous manifestations, but the skin findings are likely to be lacy and maculopapular.

Infection with SARS-CoV-2 is the most likely viral cause of the hemorrhagic lesions in this patient. Cutaneous manifestations of Covid-19 include morbilliform, pernio, or chilblain acral lesions, as well as livedo reticularis, urticaria, vesicles, and purpuric and hemorrhagic lesions.² The purpura associated with Covid-19 is probably related to the prothrombotic effects of viral infection of endothelial cells. However, this patient's rash was targetoid in nature and not the diffuse purpura that is typically described with SARS-CoV-2 infection.

BACTERIAL INFECTIONS

Many bacteria can cause skin lesions and rashes. In this patient, blood cultures were transiently positive for *S. pneumoniae*. Pneumococcus has a polysaccharide capsule that is capable of evading the immune response, and this bacterial pathogen is the most common cause of pneumonia and meningitis in adults. Infection with *S. pneumoniae* has a breadth of cutaneous manifestations, which range from macular lesions to rapidly progressive vascular-bed processes, such as purpura fulminans.³ Purpura fulminans results from the rapid evolution of microthrombi in the capillary bed, which leads to large areas of ischemia. This manifestation can also be associated with other aggressive bacterial infections, including *Neisseria meningitidis* and capnocytophaga infections. Although the skin lesions observed in this patient would be consistent with purpura fulminans, the pneumococcal bacteremia was brief and unlikely to have resulted in such a profound clinical presentation.

The patient had been hospitalized for almost 2 weeks, which placed him at risk for nosocomial bacterial skin infections. He was at high

risk for infection with gram-positive pathogens (such as *Staphylococcus aureus*) as well as gram-negative pathogens (such as pseudomonas), both of which can result in various skin lesions. *S. aureus* is classically associated with carbuncles, abscesses, and erysipelas. It has also been linked to vasculitic lesions that are hemorrhagic and thought to be caused by immune-related phenomena. Pseudomonas is a gram-negative rod that commonly infects immunocompromised hosts. It can result in a variety of cutaneous lesions, including ecthyma gangrenosum, which is characterized by red macules with a hemorrhagic center and often by ulceration. However, both staphylococcus and pseudomonas are easily detected on culture. The absence of these bacteria in routine blood cultures and in the skin-biopsy specimen lowers the likelihood of these infections.

FUNGAL INFECTIONS

This patient's history of kidney transplantation increases his risk of disseminated fungal infection. The control of fungal pathogens requires intact, innate, and adaptive immunity. The patient had received immunosuppressive medications for the prevention of renal allograft rejection, which increased his risk of infection with invasive fungal pathogens, including cutaneous infection.^{4–6} Some fungal pathogens can cause pulmonary infection with dissemination, which would be consistent with this patient's clinical presentation.^{5,7}

Histoplasma and blastomyces are dimorphic fungi that grow as mold in the environment and revert to the yeast phase upon infection of a human host. It is possible to find both pathogens in New England. Exposure to blastomyces can result in a severe pulmonary infection that leads to critical illness, and nearly one fifth of affected patients have dissemination to the skin, with isolated nodular lesions. Similarly, exposure to histoplasma can result in pulmonary infection with dissemination to multiple organs, including the skin; raised nodules with ulceration have been described. Although either of these fungal infections can have symptoms that are consistent with this patient's illness, they commonly result in lesions that are nodular and not purpuric.

Pathogenic yeast species, such as cryptococ-

cus, can also cause pulmonary infection with dissemination in immunocompromised hosts. *Cryptococcus* is a ubiquitous yeast that is found worldwide. It causes an initial pulmonary infection that commonly disseminates to multiple organs, including the central nervous system, liver, spleen, and skin. The related skin lesions are nodular and often umbilicated but are rarely purpuric.

Finally, pathogenic mold species can result in pulmonary infection with dissemination. The molds that most commonly infect humans include *aspergillus*, *fusarium*, and *mucorales* species. Of these molds, *aspergillus* is the least likely to spread, whereas *fusarium* disseminates and can result in hemorrhagic nodules. *Mucorales* species — including *rhizopus*, *mucor*, *rhizomucor*, *cunninghamella*, *lichtheimia*, and *apophysomyces* species — are highly invasive. *Rhizopus*, specifically, expresses proteins such as spore-coat protein homologues (CotH3), which can bind to glucose-regulated protein 78 (GRP78) on endothelial cells and promote angiogenesis.^{8,9} Infection with *mucorales* species often results in severe necrotizing processes, including those involving the skin, which are caused by infarction of multiple capillary beds.

The diagnosis of infection with *mucorales* species is challenging because of the lack of antigen-specific tests, and it often requires either growth and identification in culture or ribosomal sequencing. On histopathological examination of a biopsy specimen, the presence of pauciseptate and ribbonlike hyphae that branch at wide angles may be suggestive of *mucorales* species. In addition, *mucorales* species have a predilection for skin and soft tissue; studies suggest that 10 to 14% of affected patients have skin and soft-tissue involvement at the time of the initial presentation. Moreover, mold infection, specifically infection with *aspergillus* or *mucorales* species, has been associated with SARS-CoV-2 coinfection.¹⁰⁻¹³

In this patient, who was immunocompromised after solid-organ transplantation, presented with SARS-CoV-2 infection and critical illness, and then began to have multiple hemorrhagic, non-blanching skin lesions, I was highly suspicious of an angioinvasive process. The pneumococcal bacteremia and Covid-19 appeared to be too mild and brief to account for the severity of his

clinical condition. A pulmonary mold superinfection with dissemination was considered to be the most likely diagnosis. To rapidly establish this diagnosis, skin biopsy of the hemorrhagic lesions was requested to look for fungal elements. If this test is unrevealing, 16S ribosomal RNA sequencing can be performed to assist with the diagnosis.

CLINICAL IMPRESSION AND DERMATOLOGY MANAGEMENT

Dr. Steven T. Chen: The dermatology service was consulted twice for this patient. The first dermatology consultation was to evaluate the purpuric plaque on the sacrum (Fig. 2A). Skin biopsy revealed findings consistent with pressure-induced ischemic changes of the skin; there were no signs of infection.

The second dermatology consultation was to evaluate the new lavender purpuric patch on the left side of the lower abdomen, at the site of heparin injection (Fig. 2B). There was a pink halo surrounding the purpura. The lesion did not have a retiform appearance, which decreased the likelihood of a thrombotic or vasculitic process. At this point, the most likely diagnosis was thought to be local trauma from heparin injection, given the singular and completely macular nature of the lesion and the absence of induration. However, because the patient was immunocompromised and had a prolonged illness, there were concerns about disseminated infection, so two skin-biopsy specimens were obtained for culture and pathological evaluation.

The next day, a new purpuric patch with a halo was noted on the right inner thigh. At this point, given dissemination of the skin findings at sites that were not associated with heparin injection, the most likely diagnosis was thought to be an infection, specifically a disseminated angioinvasive fungal infection, such as mucormycosis.

CLINICAL DIAGNOSIS

Mucormycosis.

DR. MICHAEL K. MANSOUR'S DIAGNOSIS

Mucormycosis.

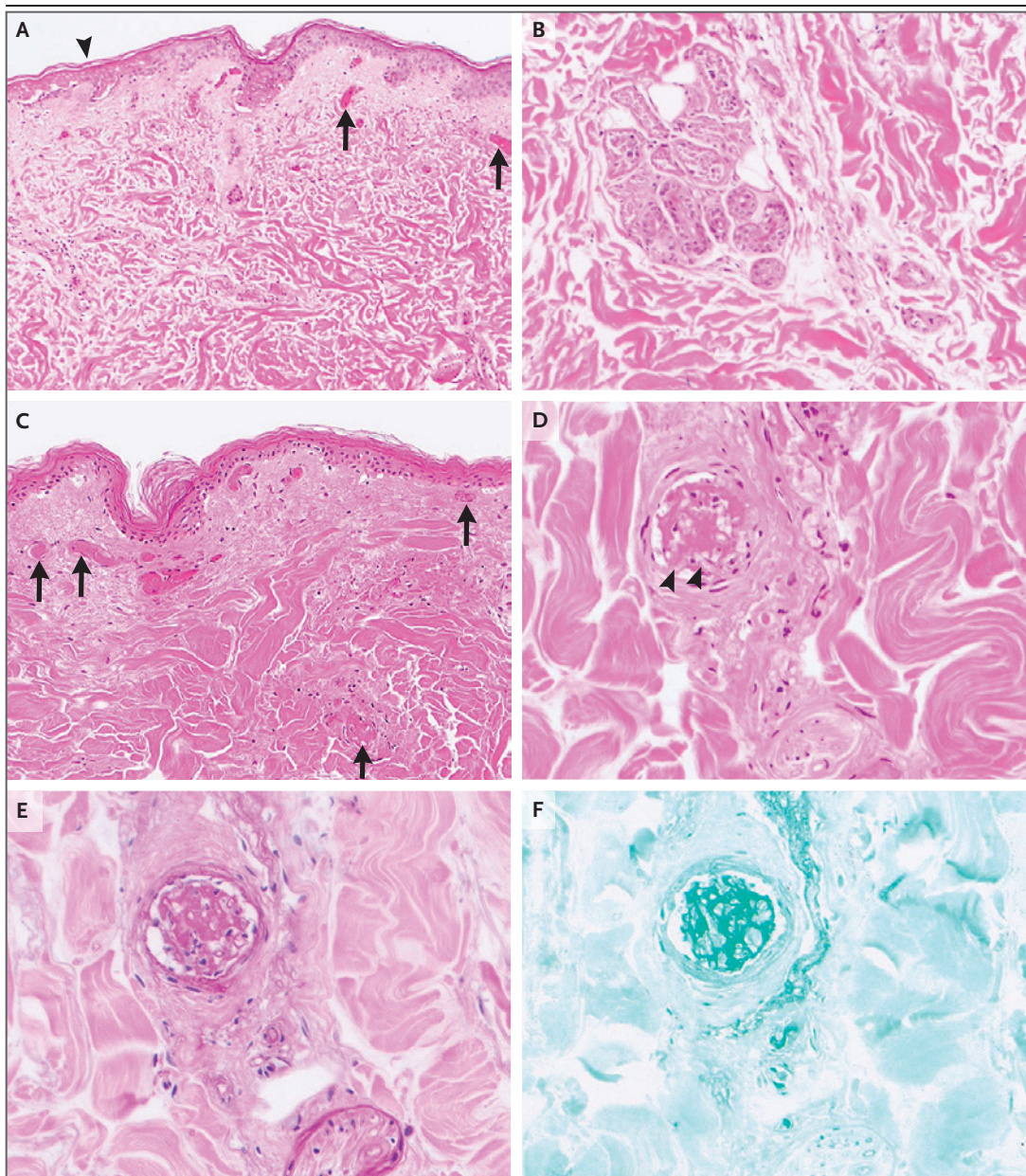


Figure 3. Skin-Biopsy Specimens.

Hematoxylin and eosin staining of a section of a biopsy specimen obtained from the left buttock (Panel A) shows epidermal necrosis (arrowhead) and vascular congestion involving superficial dermal vessels (arrows); no pathologically significant inflammatory infiltrate is visible. At higher magnification (Panel B), eccrine-gland necrosis is present, a finding consistent with pressure-induced ischemic change. Hematoxylin and eosin staining of a section of a biopsy specimen obtained from the left side of the lower abdomen (Panel C) shows a pauci-inflammatory thrombotic vasculopathy (arrows). At higher magnification (Panel D), a rare reticular dermal vessel with subtle, broad, aseptate angioinvasive fungal hyphae is present (arrowheads). Periodic acid–Schiff staining with diastase digestion (Panel E) shows fungal hyphae involving a thrombosed dermal vessel, a finding known as mucor thrombus. Grocott methenamine–silver staining (Panel F) is negative, which is consistent with cutaneous mucormycosis.

PATHOLOGICAL DISCUSSION

Dr. Rosalynn M. Nazarian: Punch biopsy of skin on the left side of the lower abdomen and on the

left buttock was performed on hospital day 14 (Fig. 3). Routine histologic evaluation of sections of the biopsy specimens revealed a pauci-inflammatory thrombotic vasculopathy with angioinva-

sive fungal hyphae and necrosis of the epidermis and eccrine glands. SARS-CoV-2 immunostaining was negative. Acid-fast, Fite, Brown–Hopps, and Grocott methenamine–silver staining was negative. Periodic acid–Schiff staining with diastase digestion showed broad fungal hyphae involving thrombosed dermal vessels, a finding that is known as mucor thrombus and is consistent with cutaneous mucormycosis. Microbiologic cultures were negative, but aseptate hyphae were identified in a fungal wet preparation, which supported the histopathological findings. Genomic sequencing of a tissue specimen later identified *Rhizomucor miehei*.

Intravascular thrombosis is a hallmark histopathological feature of cutaneous mucormycosis. This manifestation has been reported to result from platelet activation through platelet receptor FcγRIIa (the low-affinity receptor for the constant fragment of IgG) by fungal spores; the interaction of fungal spores with plasma IgG leads to the formation of immune complexes, which induces platelet aggregation.¹⁴ The epidermis and eccrine glands are metabolically active regions of the skin, which makes them sensitive to ischemia.¹⁵ In this patient, the necrosis observed on the first biopsy was probably related to hypoxia at pressure areas (a manifestation frequently described as coma bullae), and the necrosis observed on the second biopsy was probably related to infarction due to intravascular thrombosis.¹⁵

No evidence of vasculitis (vessel-wall inflammation) was observed on either skin biopsy; this helps to rule out the possibility of recurrent ANCA-associated vasculitis in this patient.¹⁶ No pathologically significant perivascular or interstitial inflammatory infiltrate was present in the skin-biopsy specimen obtained from the left buttock; this suggests the possibility of thrombotic vasculopathy related to warfarin use (warfarin-induced skin necrosis), heparin injection (heparin-induced thrombocytopenia and thrombosis), disseminated intravascular coagulation, or an inherited procoagulant state, such as antithrombin III deficiency.¹⁷ Covid-19 can also induce a prothrombotic state that is associated in part with abnormal levels of acute-phase reactants (e.g., elevated D-dimer levels and decreased antithrombin levels) that contribute to thrombotic events in affected patients.¹⁸ The detection of angioinvasive fungal forms helps to distinguish cutaneous mucormycosis from these histopatho-

logical mimics. Because mucor thrombus may be subtle, vigilance is needed to avoid a missed diagnosis.

PATHOLOGICAL DIAGNOSIS

Cutaneous mucormycosis.

DISCUSSION OF MANAGEMENT

Dr. Jay A. Fishman: Susceptibility to infection is determined by the interplay between epidemiologic exposures and the “net state of immunosuppression” in the host, including the immunosuppressive regimen, preexisting immunity, genetics of the immune response, coexisting conditions, and immunomodulatory coinfections, such as CMV or SARS-CoV-2 coinfection.^{19–21} Transplant recipients commonly have a combination of deficits (e.g., neutropenia, viral coinfection, or T-lymphocyte dysfunction) and coexisting conditions (e.g., diabetes, kidney or heart failure, antimicrobial exposures, or a history of complex surgery with nosocomial exposures).^{19,20,22,23} In this patient, immunosuppressive therapy and lymphocyte depletion posed an increased risk of infection.

For invasive mold infection, transplantation-associated risk factors include renal replacement therapy, blood loss or transfusion of more than 40 units, reoperation, a prolonged stay in the ICU, CMV coinfection, previous mold colonization, and critical illness at the time of transplantation.^{24–29} Diabetes, particularly with diabetic ketoacidosis, is a commonly associated risk factor. Liver transplantation alone is associated with a risk of mucormycosis that is five times as high as the risk in the general population.³⁰ Previous exposure to voriconazole can confer a predisposition to mucormycosis.³¹ The skin (e.g., the site of a chest tube, burn, trauma, or surgery) is the portal of entry in up to 56% of cases of mucormycosis, often in immunocompromised hosts.^{32,33} Allograft-derived mucormycosis has been reported, most often in lung-transplant recipients, and related skin lesions are common.^{33–35}

Mucormycosis is a rapidly progressive disease in transplant recipients. To the degree possible, a reduction in immunosuppression would be considered. The reduction or elimination of glucocorticoid therapy can improve neutrophil function and the control of diabetes. Such a reduction may result in an increased risk of graft

rejection; the replacement of glucocorticoids with calcineurin inhibitors may result in an increased risk of nephrotoxicity in kidney-transplant recipients. Imaging and biopsy are performed immediately to confirm the location and extent of the disease, followed by excision of infected tissues to clean the margins.^{36,37} Early antifungal therapy is warranted.^{38,39} In a multicenter, retrospective study involving patients with mucormycosis who were treated with lipid formulations of amphotericin B (mainly liposomal amphotericin B), the probability of survival was 0.72 among the 53 patients who started treatment within 3 days after diagnosis, as compared with a probability of only 0.33 among the 6 patients who started treatment beyond day 3.⁴⁰

In a small, retrospective analysis of published reports of mucormycosis in kidney-transplant recipients, survival was 70.2% after surgical débridement and treatment with an antifungal agent, as compared with 36.4% after surgery alone and 32.4% after antifungal treatment alone.⁴¹ A separate analysis of only the antifungal treatment received (without note of whether surgical débridement occurred) showed a survival of 92.3% with posaconazole, 73.4% with liposomal amphotericin B, and 47.4% with amphotericin B deoxycholate. Other therapies have been less effective.^{30,42} Prospective clinical trials investigating primary treatment for mucormycosis have focused on amphotericin B and isavuconazole.^{43,44} Amphotericin B has variable mucorales-specific activity in vitro.⁴⁵ Because transplant recipients who are receiving calcineurin inhibitor therapy for immunosuppression have renal impairment, lipid formulations of amphotericin B are the preferred initial treatment for mucormycosis.

The duration of therapy is defined on an individual basis. It is helpful to have “something to follow,” such as imaging-based or clinical

evidence that the infection has resolved. A duration of at least 6 to 8 weeks is common, and maintenance therapy may be indicated in patients with persistent immunosuppression.

MUCORALES AND COVID-19

Dr. Fishman: Immune deficits are exacerbated in patients with SARS-CoV-2 infection.⁴⁶ The risk associated with these effects is amplified in critically ill patients in the ICU, in whom nosocomial exposures and additional immunosuppression (e.g., use of dexamethasone) increase the risk of infection due to aspergillus or mucorales species.⁴⁷ The increased risk of fungal infection may reflect a diminished capacity for monocyte phagocytosis and killing, which has been observed with CMV infection.²¹ Coexisting conditions, such as diabetes and heart failure, have been associated with ICU admission and death in patients with Covid-19. Mucormycosis has been observed in patients with Covid-19 in the absence of diabetes, intubation, and ICU care; this factor possibly emphasizes the role of exogenous immunosuppression in the pathogenesis of the infection.⁴⁸

Dr. Mansour: Treatment with liposomal amphotericin B was continued. Given the critical nature of this patient's illness and the poor prognosis associated with extensive dissemination of angioinvasive mucormycosis, a strategy of comfort measures only was selected by the patient's family. The patient died shortly thereafter.

FINAL DIAGNOSIS

Mucormycosis.

This case was presented at the Internal Medicine Comprehensive Review and Update 2022, directed by Drs. Joshua Metlay, Rocio Hurtado, and David M. Dudzinski.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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